

AI Engineer Roadmap - Study Notes (Volume 1)

These notes are designed to accompany your daily learning roadmap. They focus on understanding concepts rather than memorizing code.

Chapter 1: Artificial Intelligence

- What is AI?

AI is the broad field of building systems that perform tasks requiring human intelligence.

- Machine Learning

Models learn patterns from data instead of being explicitly programmed.

- Deep Learning

A subset of ML using neural networks with multiple layers.

- Key Terms

Features, Labels, Model, Prediction, Training, Inference.

Chapter 2: Neural Networks

- Weights determine the importance of each input.
- Bias shifts the output independently of the inputs.
- Loss Function measures prediction error.
- Backpropagation computes gradients.
- Optimizer updates weights and bias to reduce loss.
- Epoch = one complete pass through the training dataset.

Chapter 3: Python Essentials

Topics:

- Variables and Data Types
- Lists
- Tuples
- Sets
- Dictionaries
- Loops
- Functions
- Strings
- List Comprehensions

Chapter 4: Functions

Remember:

- `print()` displays output.
- `return` sends a value back and immediately ends the function.

Important interview topics:

- Parameters
- Arguments
- Default Parameters
- Keyword Arguments

- Scope

Chapter 5: Strings

Practice:

- Reverse a string
- Count vowels
- Check palindrome
- Character frequency
- Word frequency

Chapter 6: Interview Coding Patterns

1. Maximum/Minimum
2. Frequency Counter (Dictionary)
3. Two Variables Tracking (Largest/Second Largest)
4. Traversal Pattern
5. Two Pointer Technique (coming later)

Future Volumes

Volume 2:
NumPy, Pandas, Data Visualization

Volume 3:
Machine Learning

Volume 4:
Deep Learning with PyTorch

Volume 5:
Computer Vision

Volume 6:
Large Language Models

Volume 7:
MLOps & Deployment